

BevOS

Team Silver

Design Presentation

June 2026

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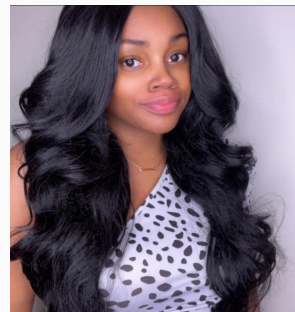
Team Bios

Team Silver

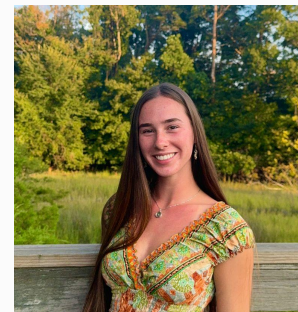
BevOS



Phillip De Guzman



Amber Ponder



Jeweliana Beayon



Julien Fernandes



Matt Kirby

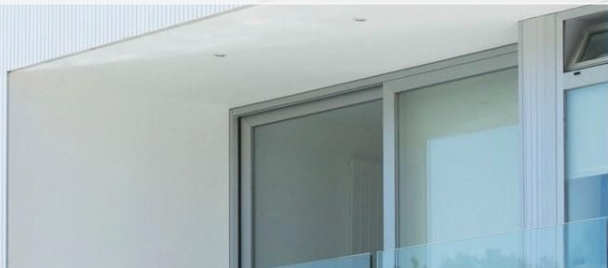


Kushal Honnappa
Mentor

Elevator Pitch

The beverage packaging industry was estimated to be at USD 157.73 billion in 2023 and is projected to reach USD 222.08 billion by 2030 (Grand View Research, 2024.) Beverage manufacturers often manage product launches through disconnected spreadsheets, email chains, and multiple software systems. This fragmented approach can lead to delays, communication breakdowns, and costly operational errors.

BevOS is a centralized workflow management platform designed specifically for beverage manufacturing. The platform connects suppliers, co-packers, logistics providers, and operational teams into a single system, providing visibility across the entire production lifecycle. By streamlining approvals, tracking production activities, and coordinating logistics, BevOS helps companies operate more efficiently and scale their business with fewer operational bottlenecks.



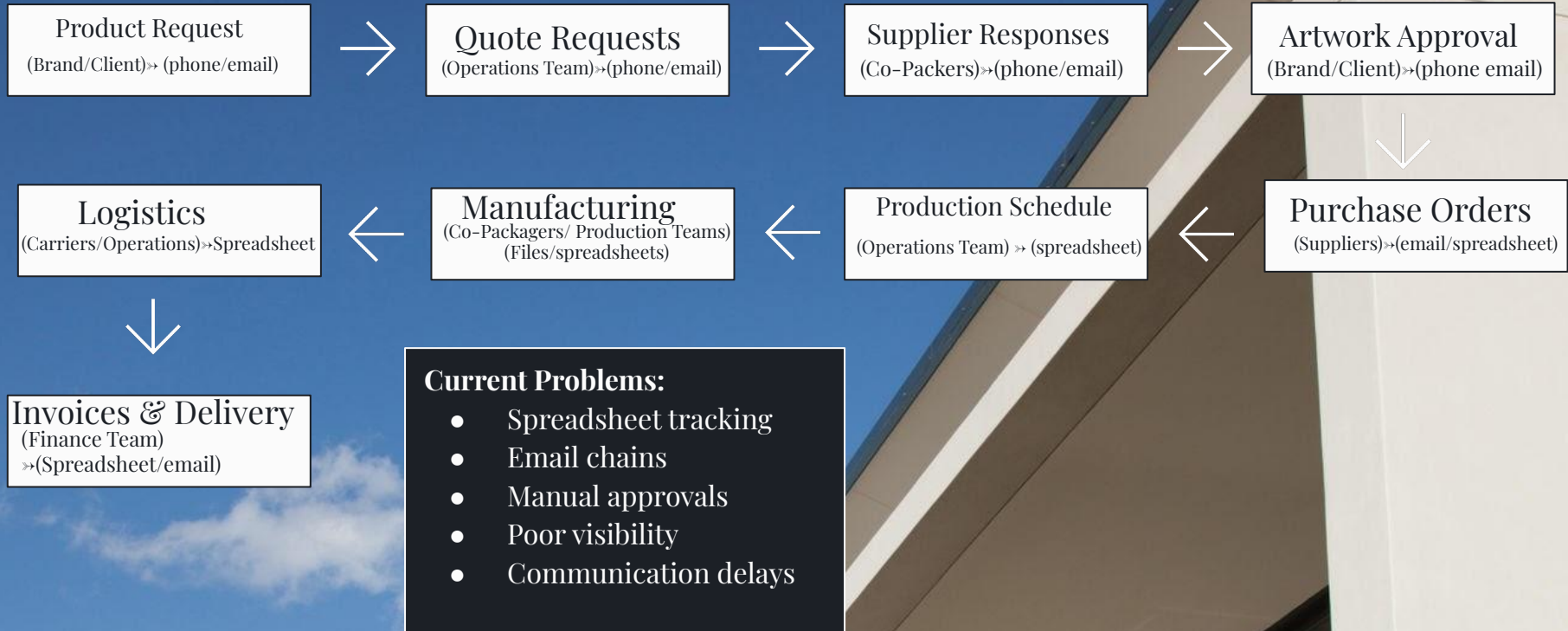
The “Problem”



As beverage companies grow, coordination of production activities become more challenging to manage. The beverage manufacturing industry relies on multiple spreadsheets, email chains, and phone calls. While these are all helpful tools that allow companies to operate they create inefficiencies as a business grows. Important updates can become lost in long email chains, causing delays and miscommunication between brands, suppliers, co-packers, and logistic providers. Approval processes for artwork, purchase orders, and production schedules frequently require a manual follow up creating bottlenecks that slow projects and increase workload.

BevOS addresses these challenges by providing a centralized workflow management platform specifically design for beverage manufacturing operations.

Current Process Flow



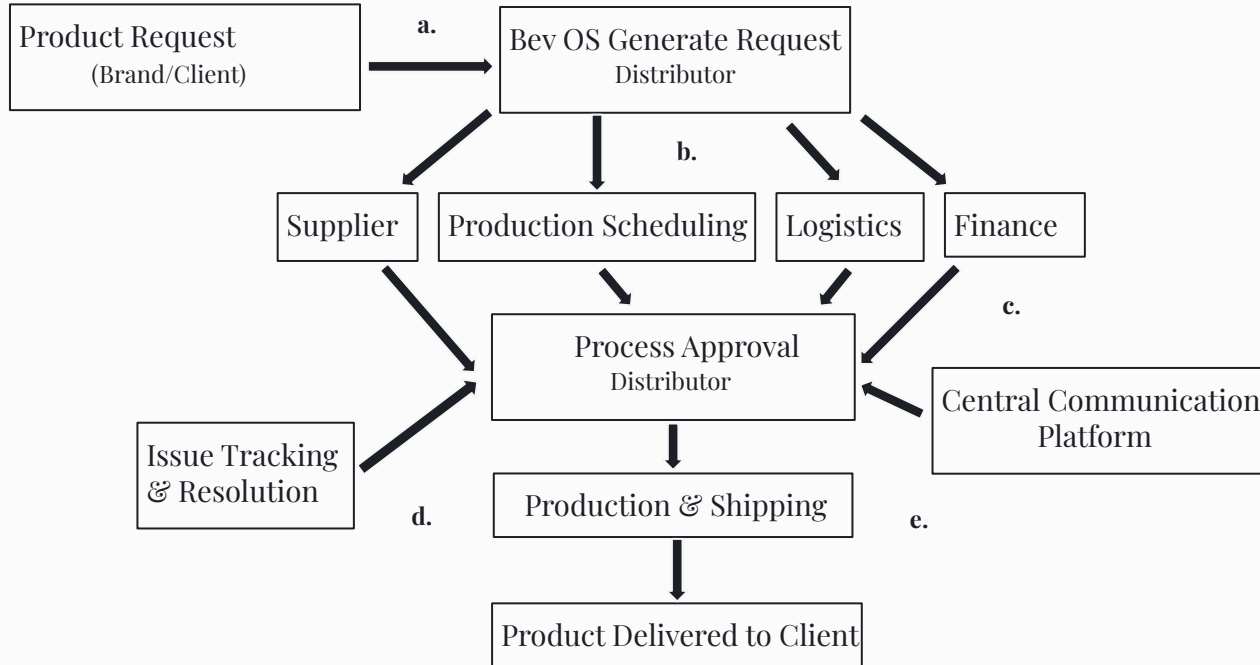


The Solution Statement

Current beverage manufacturing operations rely heavily on manual coordination between suppliers, production facilities, logistics providers, and finance teams. These processes are often managed through spreadsheets, emails, and disconnected software applications, creating inefficiencies and increasing the risk of errors.

BevOS addresses this problem by providing a unified operational platform that centralizes workflow management, approval tracking, production scheduling, document management, and logistics coordination. The goal is to improve visibility, reduce manual effort, and create a scalable system that supports business growth.

Solution Process Flow



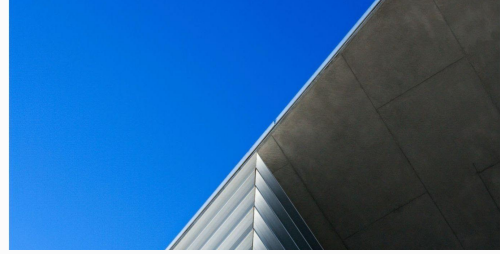
- a. Automated product request/artwork approval with escalation to partial automation when needed.
- b. Automated quote calculation with partial automation of negotiation where needed.
- c. Automated placement and follow-up on purchase orders, supplier responses, etc. on behalf of internal clients.
- d. Partial automation of internal user feedback into issue tracking to identify and consider emerging shortcomings.
- e. Repetition and tedium eliminated last step of solid plan to on store shelves fast, painless, for all involved now through minimizing communication delay, opacity, needs for manual approval, data fragmentation

What BevOS Will Do



- Centralize operational workflows
- Improve communication between stakeholders
- Track quotes, approvals, and purchase orders
- Manage production scheduling and tracking
- Coordinate logistics activities
- Store and organize operational documents
- Provide visibility through dashboards and reporting

Enterprise Resource Planning



Enterprise Resource Planning (ERP) is a software system that integrates and manages core business processes such as finance, inventory, manufacturing, sales, and HR all with one platform. ERP provides a centralized database for unified workflow that allows multiple people to share information across departments.

What BevOS Will NOT Do



- Physically manufacture products
- Operate production equipment
- Replace all existing ERP systems
- Handle direct consumer sales
- Manage warehouse automation systems
- Replace human decision-making processes

Development Tools



Testing



Source control



Integrated Development
Environment



Framework



GitHub Actions
CI/CD

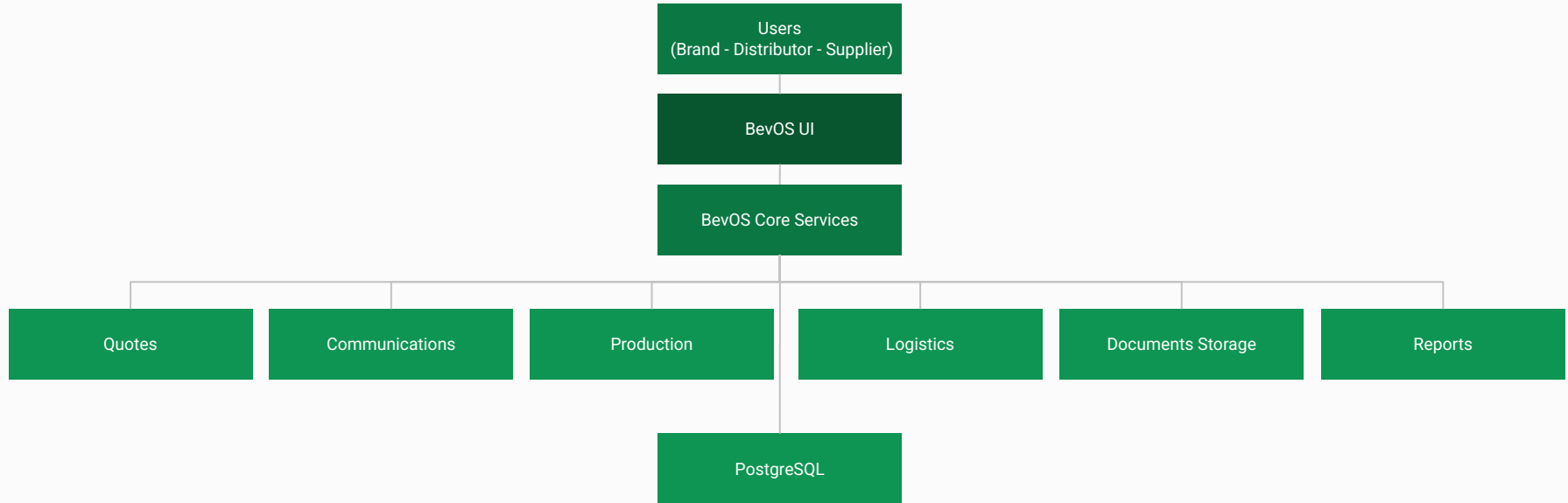
Major Functional Components



Main language



Major Functional Components Diagram (MFCD)



Client Risks



		Impact				
		Very Low (1)	Low (2)	Medium (3)	High (4)	Severe (5)
Severe (5)						
High (4)					CR3 CR2	CR5
Medium (3)					CR4	CR1
Low (2)				CM1		
Very Low (1)						

CR1: Breach of Client Confidentiality/IP

CR2: Adoption Friction and Production Error

CR3: Centralized Service Outages

CR4: Wrong Information

CR5: Customer Trust

CM1: Role-based access controls with only need to know data provided break up single point of failure potential. Sharding.

CM2: Phased adoption through trial simulation, real world process modeling, minimum workflow adoption, full adoption minimize potential losses and rapid issue discovery/resolution.

CM3: Local redundancy in infrastructure alongside minimizing need for communication with central database provides tolerance for partial functionality during minor outages.

CM4: System design emphasizes cohering with reality as provided by the user but prevents entry of known nonsensical data, validation falls to user facilitated by system metrics.

CM5: Emphasis on transparent and continuing dialogue between development and domain clients gradually handing off to client tools to self-maintain workflows.

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CM1: Role-based access control, Workflow & Database design flexibility for hierarchical client defined confidentiality.

CM2: Phased adoption through trial simulation, real world process modeling, minimum workflow adoption, full adoption after domain influence on design and growth is ingested.

CM3: Local redundancy in infrastructure alongside minimizing need for communication with central database provides tolerance for partial functionality during minor outages.

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Technical Risks



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High (4)			TR3		TR4
Medium (3)		TM3	TM1 TM4		TR2
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TR1: Domain & Process Scalability

TR2: Cybersecurity

TR3: Opaque Automation

TR4: Accumulating Technical Debt

TM1: Limited and gradual increase in system scope leaning towards strong foundation which clients and developers can build on to gradually encompass given process domain. Continual domain feedback and dialogue.

TM2: Separation of process coordination network except through specific role interactions. Encryption and local emphasis on sensitive data storage.

TM3: Data and metric visualization functionality which emphasizes multiple perspectives to a clear big picture.

TM4: Recognition of and minimization of the technical debt issue as fundamental tenet of Bevos with emphasis on futureproof Careful foundation construction and gradual architecture design.

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TR2: Cybersecurity

TR3: Opaque Automation

TR4: Accumulating Technical Debt (Unmaintainable)

TR5: Long-Term Viability? (

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TR5: System as Whole vs. Parts vs. Gestalt Design

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TM5: Acknowledge the binary tensions as part of the design framework that inform all other aspects, getting to Gestalt as goal.

Legal Risks



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	Very Low (1)	Low (2)	Medium (3)	High (4)	Severe (5)
Severe (5)					
High (4)				LR2	
Medium (3)	LM2		LR3		
Low (2)		LM3		LR1	
Very Low (1)			LM1		

LR1: Regulatory Compliance

LR2: Multi-Party Contract Complications

LR3: System Client Obligations

LM1: Role-based access controls with only need to know data provided break up single point of failure potential.

LM2: Phased adoption through trial simulation, real world process modeling, minimum workflow adoption, full adoption minimize potential losses and rapid issue discovery/resolution.

LM3: System emphasizes trying to represent reality as closely as possible versus serving as source of truth, internal tooling empowers rather than replaces clients self-oversight.

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LM1: Role-based access controls with only need to know data provided break up single point of failure potential.

LM2: Phased adoption through trial simulation, real world process modeling, minimum workflow adoption, full adoption minimize potential losses and rapid issue discovery/resolution.

LM3: System emphasizes trying to represent reality as closely as possible versus serving as source of truth, internal tooling empowers rather than replaces clients self-oversight.

Competition Matrix

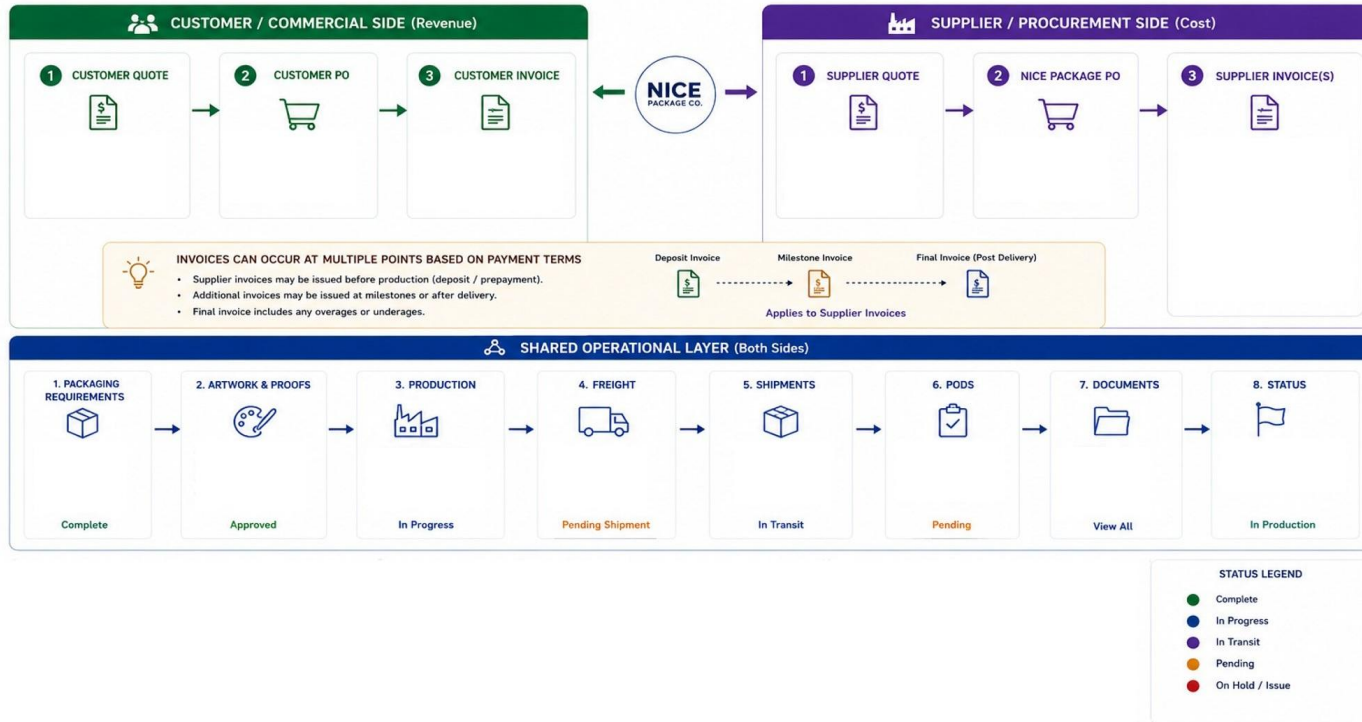
Feature	Email & Spreadsheets	Generic ERP	BevOS
Centralized Workflow			
Beverage Industry Focus			
Approval Tracking			
Production Tracking			
Logistics Visibility			
Document Management			

Work Breakdown Structure

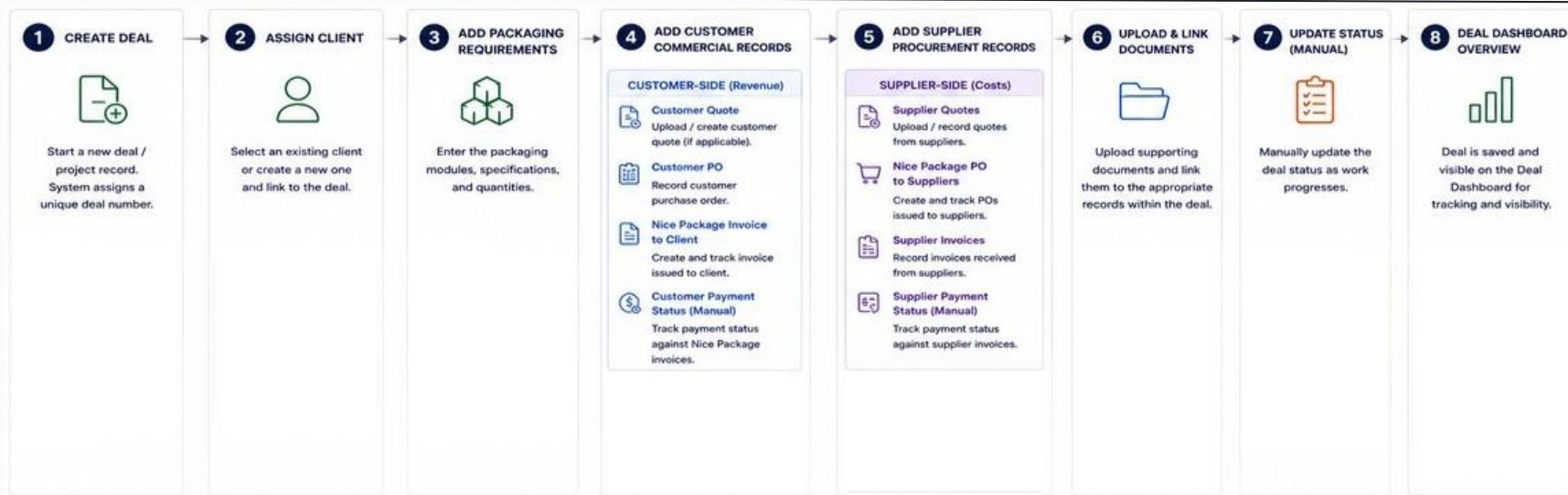


Wilson, T. (2026). BevOS Phase I Milestones [Internal company document]. Nice Package Co.

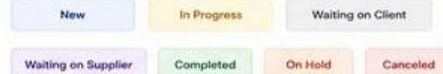
Workflow Simulation Algorithm



Deal Creation Algorithm



STATUS EXAMPLES (MANUAL)



1 Statuses are manually updated and do not trigger automations in Phase 1.

Artwork Approval Algorithm



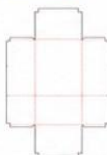
1 INTAKE & REQUIREMENTS

- Customer brief
- Branding / copy
- Colors / finish
- Barcodes / compliance
- 3D concept (if needed)



2 DIELINE / TEMPLATE

Factory provides structural dieline or template (no artwork)



3 ARTWORK CREATION

Artwork placed on dieline/template per specifications



4 PROOFS

Proofs generated for review

- Digital PDF Proof
- Color Proof (Digital)
- Color Proof (Hard Copy)
- Press Proof (if required)



5 REVIEW & APPROVALS

Internal + Customer review and sign-off

- Content
- Colors
- Barcode
- Dieline Fit
- Spelling / Compliance



6 FINAL PRODUCTION

Production files released to factory

- Print Plates
- Cutting Die
- Glue Program



Database Schema - ERD



PHASE 1 – CORE DATABASE RELATIONSHIP DIAGRAM

Contractual Phase 1 Scope Only | Internal Operational Platform

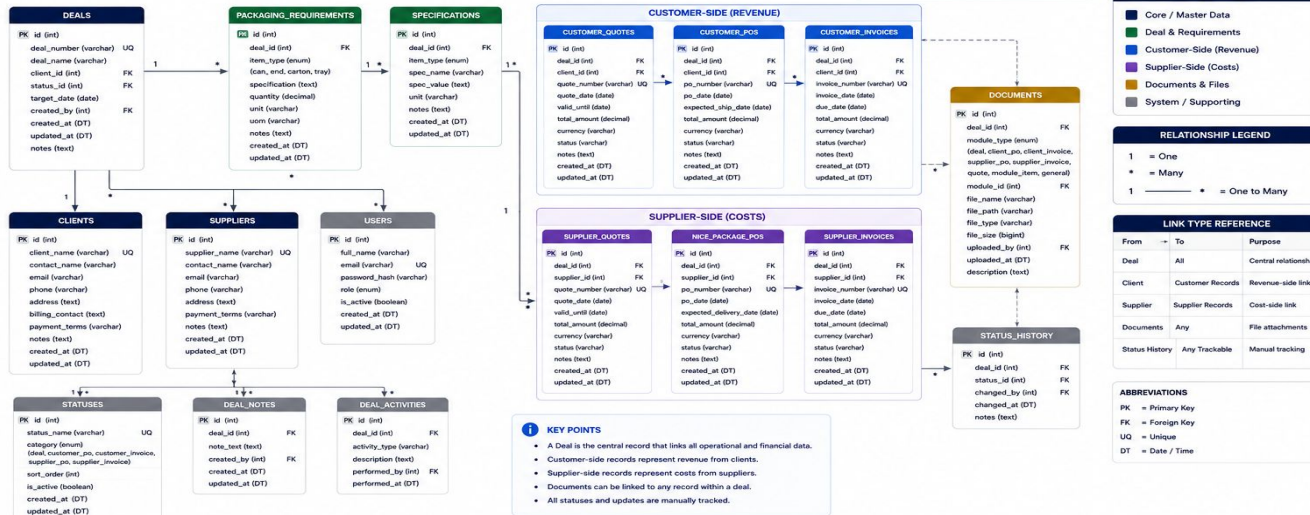
i This diagram reflects only the data structures required for Contractual Phase 1.
All data entry and tracking are manual. No automations. No external integrations.

PHASE 1 SCOPE INCLUDES

- ✓ Deal / Project system
- ✓ Client and supplier records
- ✓ Packaging requirements and specifications
- ✓ Customer-side commercial records (Quotes, POs, Invoices)
- ✓ Supplier-side procurement records (Quotes, POs, Invoices)
- ✓ Document upload and storage
- ✓ Manual status tracking
- ✓ Dashboard for visibility

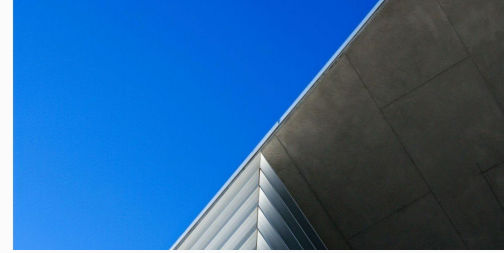
OUT OF SCOPE (PHASE 1)

- ✗ AI / OCR / Automation
- ✗ Advanced pricing / matrices
- ✗ Automated decisions / workflows
- ✗ Integrations (Xero, Cin7, Avalara, etc.)
- ✗ External user access
- ✗ Marketplace functionality



i This schema supports the complete operational tracking needs of Nice Package Co. within Contractual Phase 1.
All relationships are internal. No external system access or automations are included in Phase 1.

Database Schema - Entities



Database entities (Number of Attributes)	Deal_Activities (6)
Deals (10)	Customer_Quote (12)
Clients (11)	Customer_POS (12)
Suppliers (10)	Customer_Invoices (12)
Packaging_Requirements (10)	Supplier_Quotes (12)
Specifications (10)	Nice_Package_POS (12)
Users (8)	Supplier_Invoices (12)
Status (7)	Documents (11)
Deal_Notes (6)	Status_History (6)



Real World Product vs Prototype

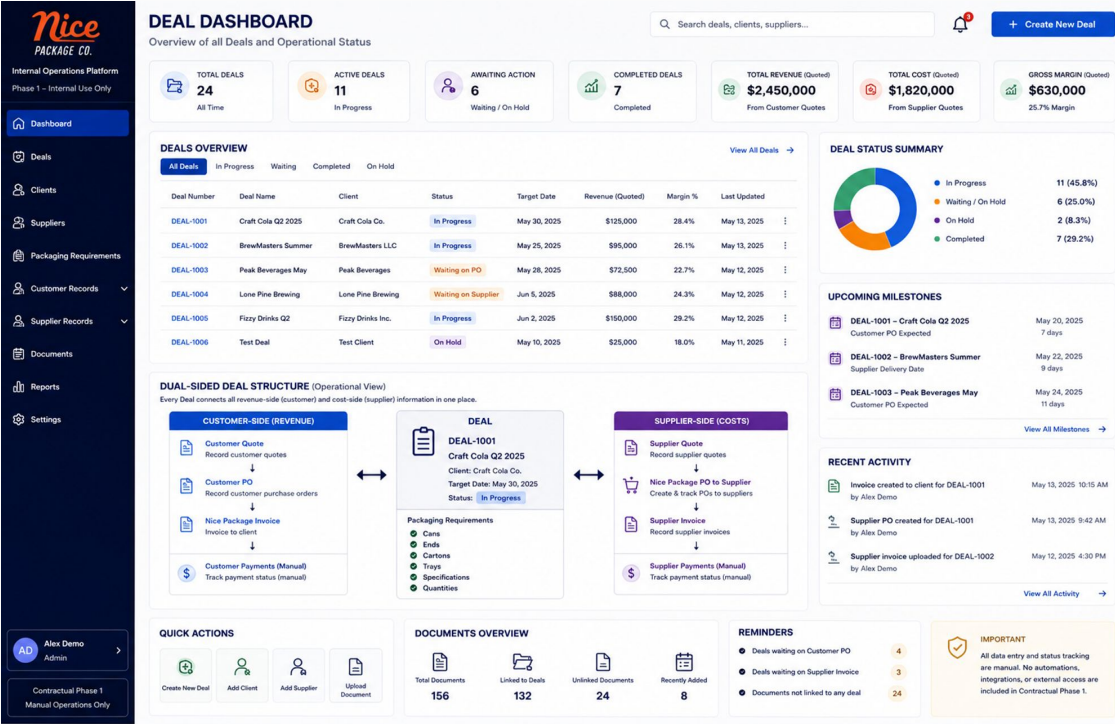
	RWP BevOS	Prototype
Centralized Workflow	Complete simulation of business	Stubs for non-target modules
Beverage Industry Focus	Initially within Adaptable Framework	Assumed sole focus
Approval Tracking	Complete	Complete
Production Tracking	Complete	Complete, drivers where needed
Logistics Visibility	Complete	Complete, synthetic data
Document Management	Technical and Client/Legal Consideration	Technical implementation only

Real World Product vs Prototype



	RWP BevOS	Prototype
Subuser Administration	Client Determined & Customizable	Proof of Concept Client Customization
Distributed Resilience	Fragmentation & Distribution Tolerance	Not considered, prototype assumes single and centralized system
Adoption Friction	Minimal Training, Cost, Risk	Focus on Training and Cost Min.
Scalability	Industry/Domain/Futureproof Flexibility & Easy Scaling	Not explored, simplest base case as focus
Security	Industry standard at least	Not explored in this prototype.
Legal	Minimum, User Support Focus	Not explored in this prototype.

GUI Mockup – Dashboard



GUI Mockup – User Management



GUI Mockup – Operations Overview



GUI Mockup – Quotes

Nice Package Co.
BevOS

Dashboard

Quotes 3

Purchase Orders 2

Production

Logistics

Documents

Settings

Travis Wilson
Ops Manager

Quotes & Approvals
PackFlow > Quotes & Approvals

Q Search...

TW

All Pending Approved In Progress Rejected On Hold

Export

QUOTE ID	CLIENT	PRODUCT	QUANTITY	VALUE	SUBMITTED	REP	STATUS	
QT-2407-001	Hydra Beverages Ltd	330ml Aluminum Can — 12-pack	500,000 units	\$84,500	Jul 18, 2026	Marcus Chen	Pending	...
QT-2407-002	Verde Juice Co.	1L PET Bottle — 6-pack	200,000 units	\$61,200	Jul 17, 2026	Sarah Okonkwo	Approved	...
QT-2407-003	Arctic Spring Water	500ml Glass Bottle	1,000,000 units	\$215,000	Jul 15, 2026	James Reilly	Approved	...
QT-2407-004	Fusion Energy Drinks	250ml Slim Can — 24-pack	750,000 units	\$130,750	Jul 14, 2026	Marcus Chen	In Progress	...
QT-2407-005	Oasis Kombucha	355ml Pouch — bulk pallet	300,000 units	\$54,900	Jul 12, 2026	Lena Fischer	Rejected	...
QT-2406-098	Cascade Craft Soda	473ml Can — seasonal edition	150,000 units	\$38,250	Jun 28, 2026	Sarah Okonkwo	Approved	...
QT-2406-095	TrueLeaf Tea	250ml Carton — 4-pack	400,000 units	\$72,000	Jun 25, 2026	Lena Fischer	On Hold	...

Showing 7 of 7 quotes

GUI Mockup – Purchase Orders

Nice Package Co.
BevOS

Dashboard

Quotes 3

Purchase Orders 2

Production

Logistics

Documents

Settings

Travis Wilson
Ops Manager

Purchase Orders
PackFlow > Purchase Orders

Q Search...

TW

+ New PO

Total Active
6

Pending Approval
2

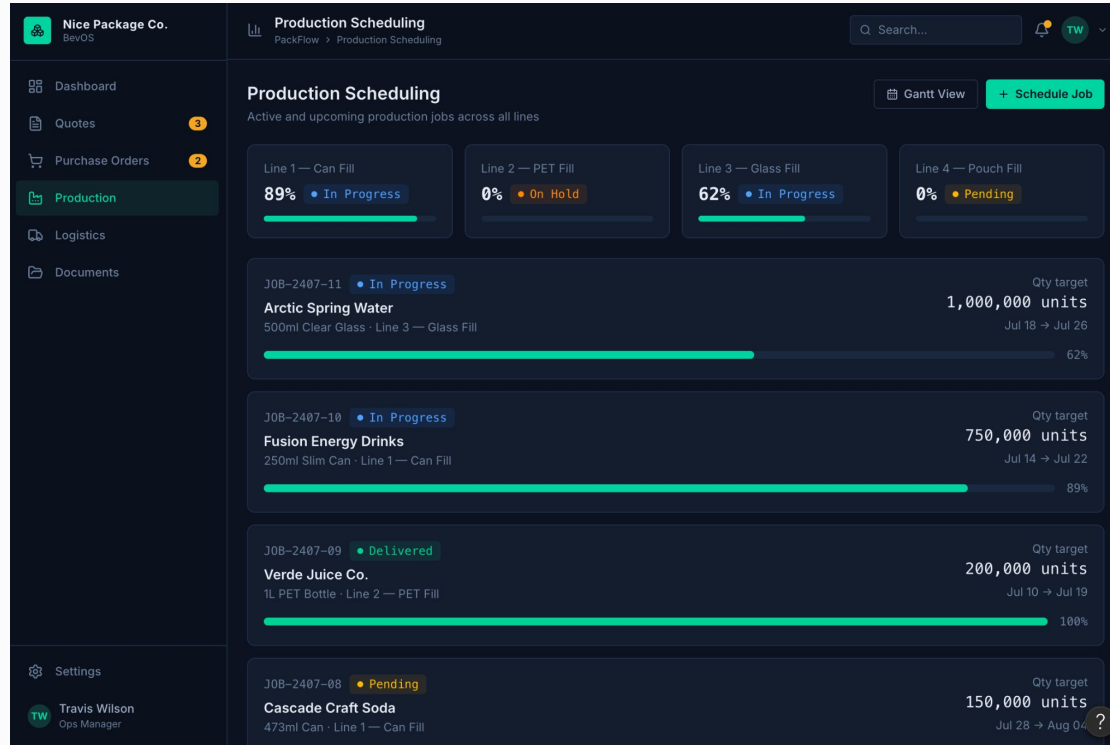
In Transit
2

Total Value
\$817.5K

PO NUMBER	SUPPLIER	DESCRIPTION	AMOUNT	ISSUED	DUE	STATUS	
P0-2407-0041	Ardagh Metal Packaging	Aluminum sheet coil — 350 MT	\$298,500	Jul 16, 2026	Aug 02, 2026	Approved	...
P0-2407-0040	Verallia Glass	Clear glass preforms — 50,000 kg	\$147,200	Jul 15, 2026	Aug 10, 2026	In Progress	...
P0-2407-0039	Coveris Flexible	Laminate foil — HPF grade, 180 rolls	\$63,400	Jul 14, 2026	Jul 28, 2026	Shipped	...
P0-2407-0038	Sun Chemical	UV-cure inks — 12 SKUs	\$31,800	Jul 10, 2026	Jul 24, 2026	Delivered	...
P0-2407-0037	Amcor Rigid Plastics	PET preforms — 28mm neck, 2M units	\$184,000	Jul 08, 2026	Aug 05, 2026	Pending	...
P0-2407-0036	Tetra Pak	Aseptic board — 250ml format, 80 rolls	\$92,600	Jul 05, 2026	Jul 30, 2026	Approved	...

?

GUI Mockup – Production Scheduling



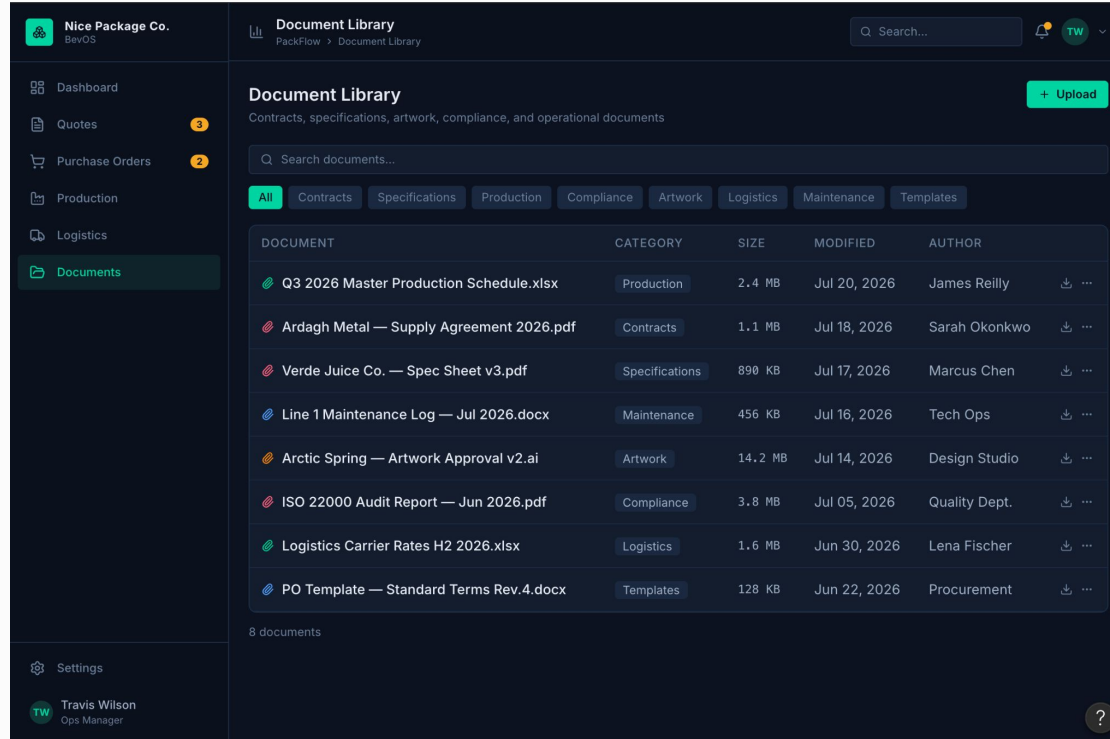
GUI Mockup – Logistics & Shipments

The mockup displays a web application interface for logistics management. The left sidebar contains navigation links: Dashboard, Quotes (3), Purchase Orders (2), Production, Logistics (selected), and Documents. The main content area is titled 'Logistics' and 'Logistics & Shipments', with a subtitle 'Inbound freight tracking and carrier coordination'. It features a 'Live shipment map' showing 5 active routes from Europe to Cork, IE. Below the map is a table listing shipments with columns for Shipment ID, Purchase Order, Carrier, Origin, Destination, ETA, Tracking, and Status.

SHIPMENT ID	PURCHASE ORDER	CARRIER	ORIGIN	DESTINATION	ETA	TRACKING	STATUS
SHP-2407-088	P0-2407-0039	DB Schenker	Leipzig, DE	Cork, IE	Jul 24, 2026	DBS-7X4920183	Shipped
SHP-2407-087	P0-2407-0038	DHL Freight	Düsseldorf, DE	Cork, IE	Jul 21, 2026	DHL-49201837X	Delivered
SHP-2407-086	P0-2407-0041	Rhenus Logistics	Hannover, DE	Cork, IE	Aug 02, 2026	RHL-2290381KA	In Progress
SHP-2407-085	P0-2407-0040	Geodis	Cognac, FR	Cork, IE	Aug 10, 2026	GE0-88120044	Pending
SHP-2407-084	P0-2407-0036	Kuehne+Nagel	Lausanne, CH	Cork, IE	Jul 30, 2026	KN-5510390EX	In Progress

The interface also includes a search bar, a filter button, and a user profile for Travis Wilson, Ops Manager.

GUI Mockup – Documents



Conclusion



The beverage manufacturing industry continues to rely on spreadsheets, email chains, manual approvals, and disconnected software systems to coordinate intricate production workflows. The spread out process causes communication delays, limited visibility, and more workload making it difficult to scale the company.

BevOS addresses all of these issues by having a centralized workflow management platform that connects brands, suppliers, co-packers, logistics, and more through a single system. Through automate workflow, real time updates, and approval management BevOs makes operations across the entire production more efficient.

References

1. Wilson, T. (2026). BevOS Presentation / Nice Package Co. Guest Presentation. CS410 Professional Workforce Development, Old Dominion University.
2. Kennedy, T. (2026). CS410 Recitation Recording and Discussion Materials. Old Dominion University.
3. Grand View Research. (2024, June). Beverage packaging market size, share & trends analysis report, 2030.
<https://www.grandviewresearch.com/industry-analysis/beverage-packaging-market>
4. Honnappa, K. (2026). Platform Visual Aid. Nice Package Co.



Appendix

1. Supporting Research
2. Diagrams
3. Additional Materials

User Stories

1. Industrial Client/Executive
2. Industrial Domain Worker/Engineer
3. Industry Partner non-Adopter
4. BevOS Development or IT Maintainer



Appendix

User Stories

1. Industrial Client/Executive

Goal: Overall elimination of inefficiencies and more tools/insights for business overall.

- Primary concern is as a secondary affected user.
- Log in see overview of business,
- Click into deals, workflows, etc. to see new perspectives on existing data to make better business decisions.

Outcome: Rather than mandating the business provide more tools, context, and insight for the business. Focus is on preserving and improving the underlying structure.



Appendix

User Stories

2. Industrial Domain Worker/Engineer

Goal: Overall less busy and more high value work in their domain / technical field.

- Primary concern as the most enmeshed within the domain.
- Log in or have continually running dashboards or pipelines into the system for their work.
- Perform work and generate metrics and feedback for further evolution.

Outcome: Important voice and design architect, the overall metric for many important aspects of design and evolution.



Appendix

User Stories

3. Industry Partner non-Adopter

Goal: Wants to continue to partner and work with BevOS adopter with as minimal costs or risk as possible, also potentially interested in being converted.

- Secondary but major concern given the prices attached to these relationships. A constant, not everyone is going to be in-network and talk the same language.
- Almost silently interface.
-

Outcome:



Appendix

User Stories

4. BevOS Development / IT Maintainer

Goal: Wants to make the business as efficient as possible while remaining flexible and not falling into the old traps of ERP and digitization efforts.

- Log in to see ticket queues, bug reports, suggestions for improvement, design arcs, domain advocate relationships, etc.
- Take in and react but also plan and move proactively to further expand platform capabilities.

Outcome: A hopeful synthesis of the system as Gestalt both Whole and Parts where it needs to be.

